Evaluation of VaR and CVaR Estimators

Using Real US Equity Return Data

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■ Historical ■ Parametric Normal ■ Student-t

Background Knowledge Sharing

Situation:
Risk managers need a single number that summarises how much an investment could lose on a bad day

2 Key Measurements:

#1 Value-at-Risk (VaR)

L = −RET (loss = - daily return), so a positive L means the stock lost money that day. The 99% VaR is the loss level that is exceeded only 1% of the time.

e.g If the 99% VaR of a stock is 4.66%, we expect a loss larger than 4.66% on roughly one trading day in a hundred.

#2 Conditional Value-at-Risk (CVaR)

Looks at how bad the losses are once the threshold has been crossed. It is the average loss on the days when the loss exceeds VaR. Formally, CVaR at the 99% level is the expected loss given that we are in the worst 1% of outcomes.

e.g. 99% CVaR is 7.26%, meaning that on the worst 1% of days the average loss is about 7.3%.

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Research Question and Study Design

Research Question:
How well do historical, Normal, and Student-t estimators recover the true 99% VaR/CVaR in finite samples, calibrated to real heavy-tailed US equity returns?

L = -RET (Daily losses = - daily returns)

Five sub-questions:

Q1 Are returns heavy-tailed? Does Student-t capture the tail?

Q2 Bias, SD, RMSE at n = 60 / 250 / 1,000 trading days?

Q3 How badly does Normal understate 99% VaR / CVaR?

Q4 Does observed exceedance match 1%? Where does Chebyshev fit?

Q5 Does performance differ: calm vs. stressed sub-periods?

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Research Question and Study Design

Step 1 — Distribution Analysis

Summary stats · JB test · Student-t MLE · Chebyshev

Purpose: To find out what the returns look like before choosing a model. We check skewness, kurtosis, and normality with the Jarque–Bera test, and fit a Student-t to see how heavy the tails are.

Step 2 — Estimator Implementation

Historical / Normal / Student-t VaR & CVaR on real data

Purpose: To compute 99% VaR and CVaR on the real returns using three approaches: Historical, Normal, and Student-t. This gives the practical numbers a risk manager would see.

Step 3 — Monte Carlo Study

n = 60 / 250 / 1,000 · 1,000 reps · seed = 42 · bias, SD, RMSE

Purpose: To measure how accurate each estimator is, which real data cannot show directly. We simulate from a known Student-t distribution, so we know the true VaR and CVaR. Then we measure bias and RMSE across 1,000 replications at n = 60, 250 and 1,000.

Step 4 — Backtesting

Rolling 250-day window · Kupiec POF test · calm vs. stressed

Purpose: To check whether the estimators behave as promised when applied to actual market returns. Each day's VaR is estimated from the previous 250 days and compared with the next day's loss, and the Kupiec test asks whether the breach rate matches 1%.

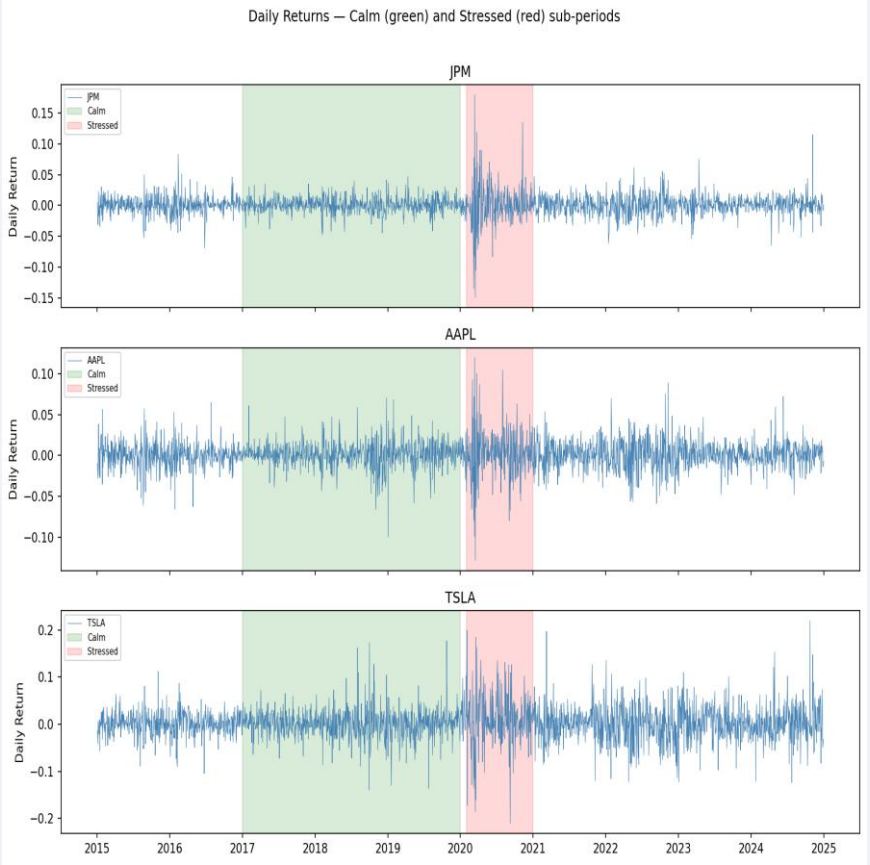
Data: WRDS CRSP Daily Returns, 2015–2024

Ticker	Name	Profile	N (days)
JPM	JPMorgan Chase	Stable blue-chip — financials	2,516
AAPL	Apple Inc.	Stable blue-chip — technology	2,516
TSLA	Tesla Inc.	High-volatility growth	2,516

Sub-periods

Period	Window	Rationale
Calm	Jan 2017 – Dec 2019	Pre-COVID low-volatility bull run
Stressed	Feb – Dec 2020	COVID crash and recovery

Source: WRDS CRSP (crsp.dsf) · Licensed data



Green shading = Calm period · Red shading = Stressed period

Returns Are Heavy-Tailed: Normality Rejected for All Tickers

JPM

13.9

excess kurtosis

JB test: p ≈ 0.000 ✗ Reject Normal

Daily SD: 1.72%

AAPL

5.3

excess kurtosis

JB test: p ≈ 0.000 ✗ Reject Normal

Daily SD: 1.79%

TSLA

4.4

excess kurtosis

JB test: p ≈ 0.000 ✗ Reject Normal

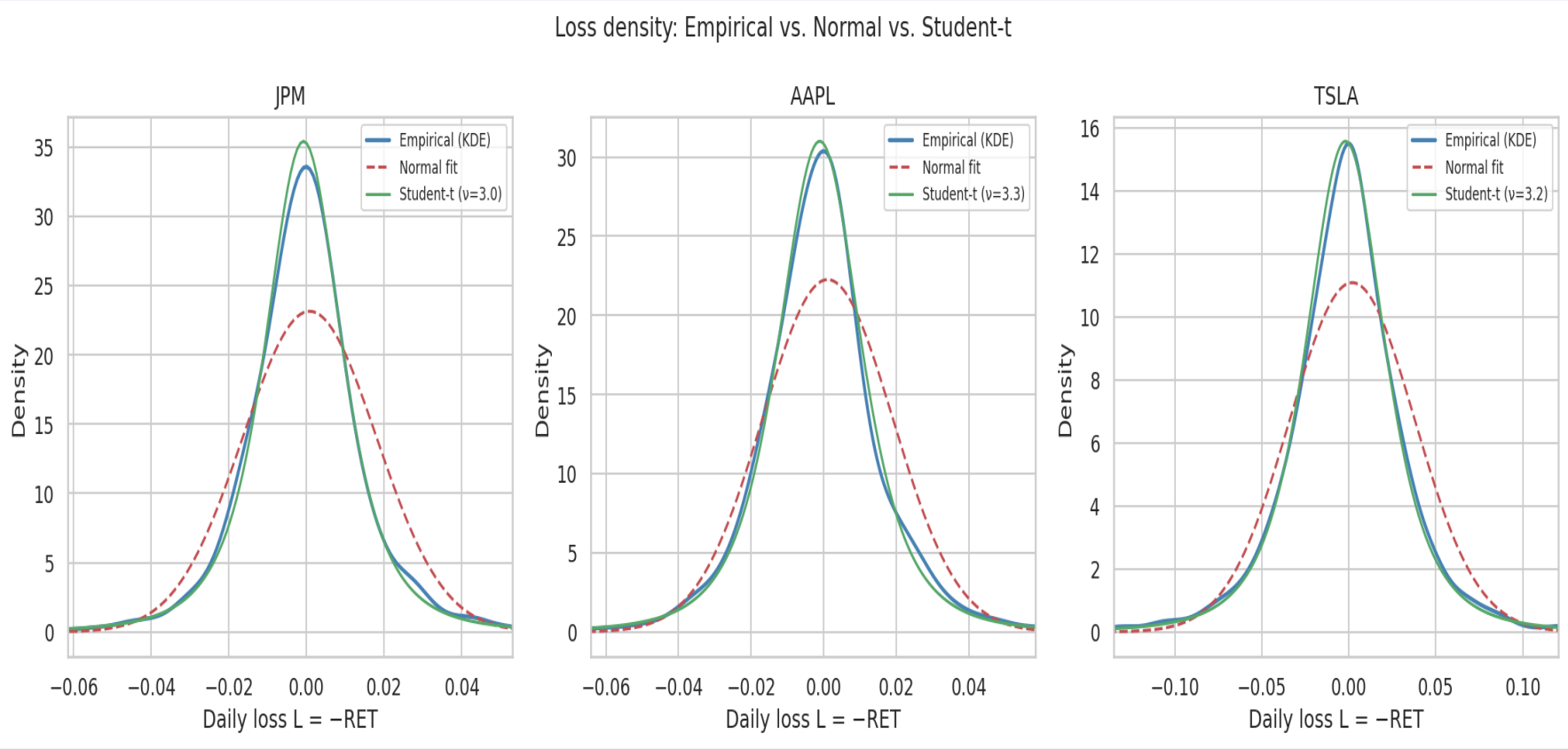
Daily SD: 3.60%

Fitted Student-t degrees of freedom (ν) by sub-period

Ticker	Full ν	Calm ν	Stressed ν	Interpretation
JPM	2.98	4.12	2.68	Heavier tails under stress — expected
AAPL	3.33	2.96	3.00	Consistently heavy tails across regimes
TSLA	3.22	3.41	3.36	Heavy tails throughout

ν → ∞ = Normal distribution · ν ≈ 3–4 confirms very heavy tails — far from Normal assumption

Student-t Tracks the Empirical Tail; Normal Misses It



■ Historical ■ Parametric Normal ■ Student-t

Chebyshev Inequality: Normal Is Systematically Optimistic in the Tails

$$P(|R - \mu| \geq k\sigma) \leq 1 / k^2$$

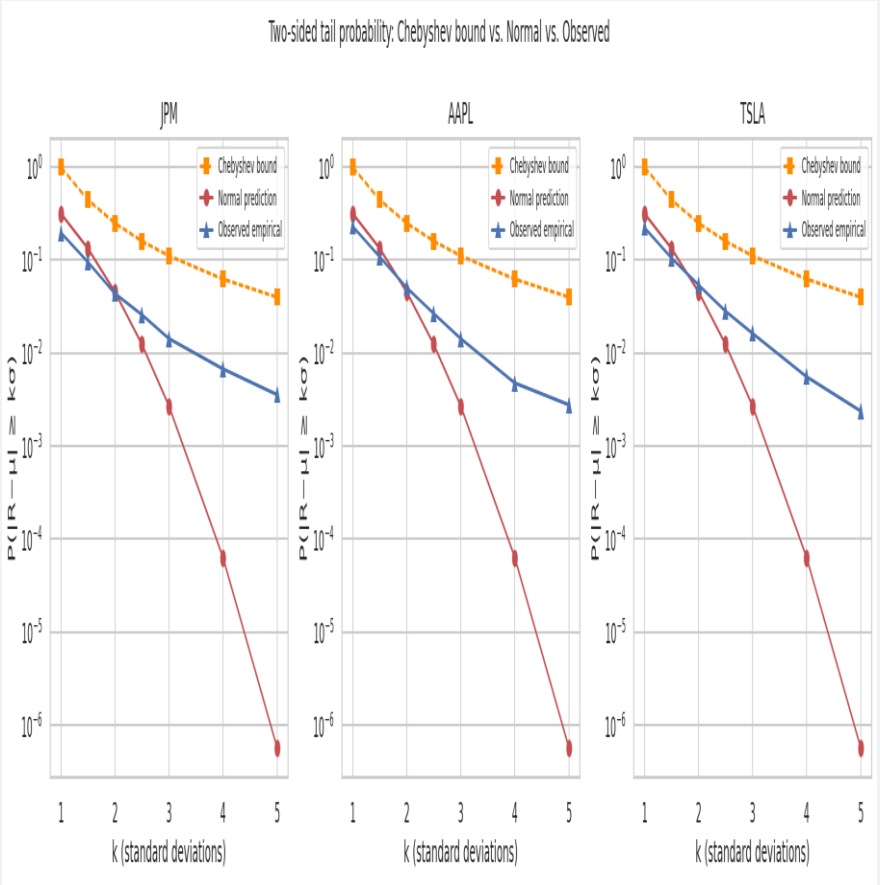
Why Chebyshev — not Markov?

Markov's inequality requires a non-negative variable. The daily return R (and loss L = −RET) can be negative, violating this condition. Chebyshev applies to any variable with finite variance and does not require non-negativity.

At k = 3:
For all tickers, Normal predicts 0.27% two-sided probability
Observed: 1.4–1.6% across all tickers (5× higher)

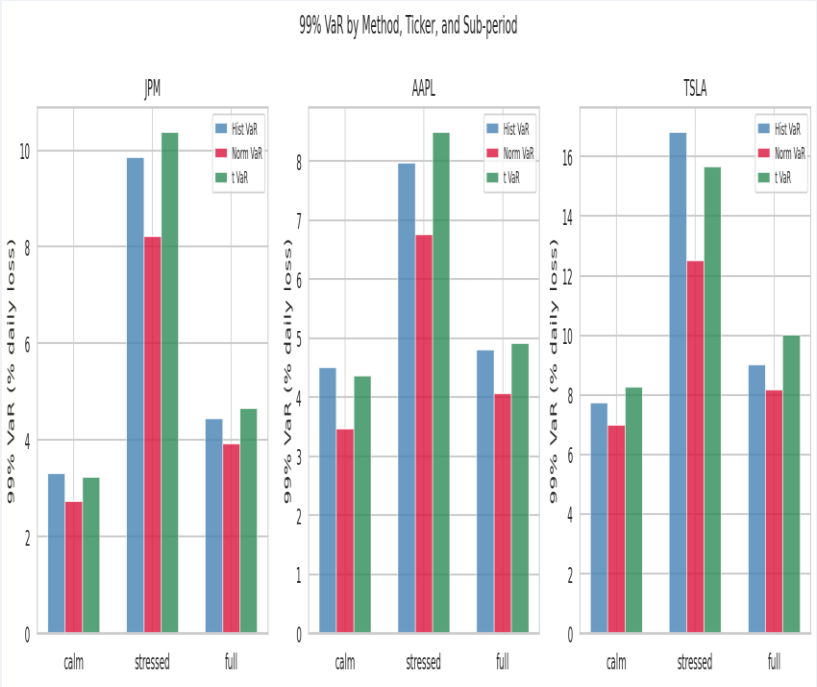
Chebyshev:
Gives 11.1% (1/9) upper bound — conservative but always valid (loose)
Distribution-free: safe regardless of tail shape

Implication:
Normal VaR underestimates probability of large losses
Chebyshev motivates moving to a heavier-tailed model



99% VaR: Normal Always Lowest; Gap Widens Sharply Under Stress

Ticker	Period	Hist VaR	Norm VaR	t VaR
JPM	Full	4.44%	3.93%	4.66%
JPM	Calm	3.31%	2.73%	3.23%
JPM	Stressed	9.85%	8.21%	10.38%
AAPL	Full	4.80%	4.06%	4.91%
AAPL	Calm	4.50%	3.47%	4.37%
AAPL	Stressed	7.98%	6.75%	8.49%
TSLA	Full	9.02%	8.18%	10.02%
TSLA	Calm	7.75%	6.99%	8.28%
TSLA	Stressed	16.82%	12.51%	15.66%



Normal VaR is consistently the lowest of the three estimators.
The gap vs. Student-t widens sharply during stressed markets.

CVaR gap is even larger: TSLA stressed — t CVaR = 23.5% vs. Normal CVaR = 14.5% (−38% shortfall)

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Step 3

Monte Carlo: Bias Converges to Zero — Except for the Normal Estimator

Calibrated Student-t DGP · 1,000 replications · Seed = 42 · n = 60 / 250 / 1,000 trading days · 99% VaR

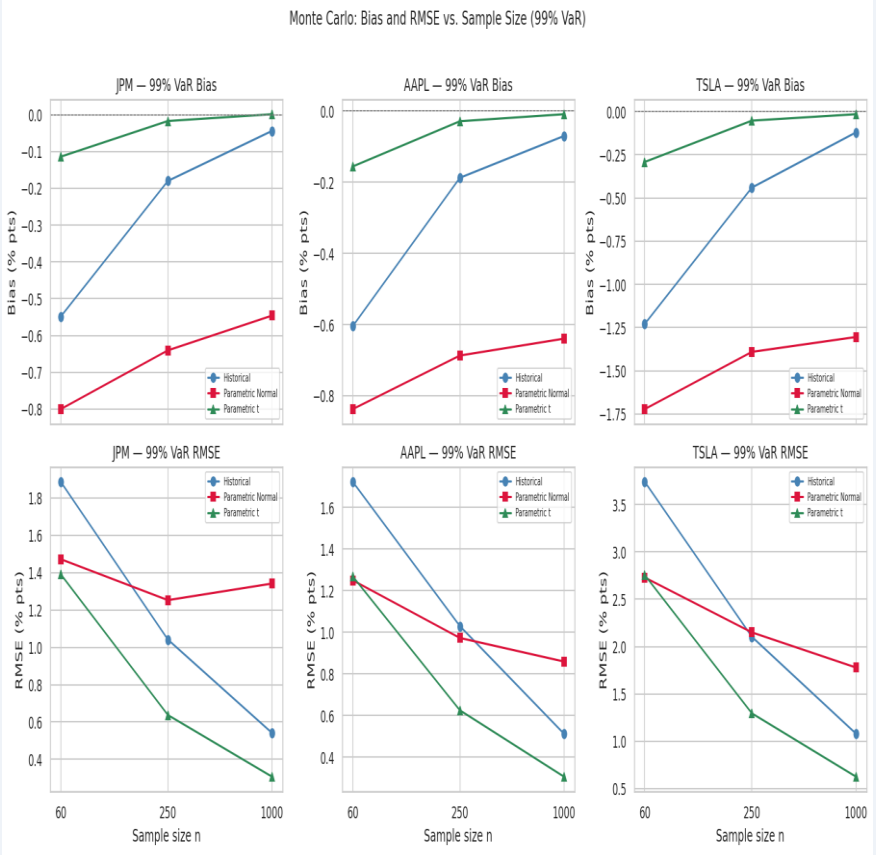
Bias at n = 1,000 days (% of true VaR)

	JPM	AAPL	TSLA
Historical	−0.9%	−1.4%	−1.2%
Normal	**−11.7%**	**−13.0%**	**−13.0%**
Student-t	≈0.0%	≈0.0%	≈0.0%

TSLA 99% VaR RMSE: n = 60 → n = 1,000

Estimator	n = 60	n = 1,000	Improvement
Historical	3.7%	1.1%	−71%
Normal	**2.7%**	**1.8%**	−35%
Student-t	2.8%	**0.6%**	−78%

The Normal estimator's bias does not shrink with more data, so it reflects a wrong model rather than limited sample size.
The other estimators' errors fall in proportion to 1/√n (follows CLT)



■ Historical ■ Parametric Normal ■ Student-t

Key Finding: Normal Understates 99% VaR by ~13%, CVaR by ~33%

JPM

−11.7%

Normal VaR bias

CVaR bias: −34.9%

AAPL

−13.0%

Normal VaR bias

CVaR bias: −32.8%

TSLA

−13.0%

Normal VaR bias

CVaR bias: −33.7%

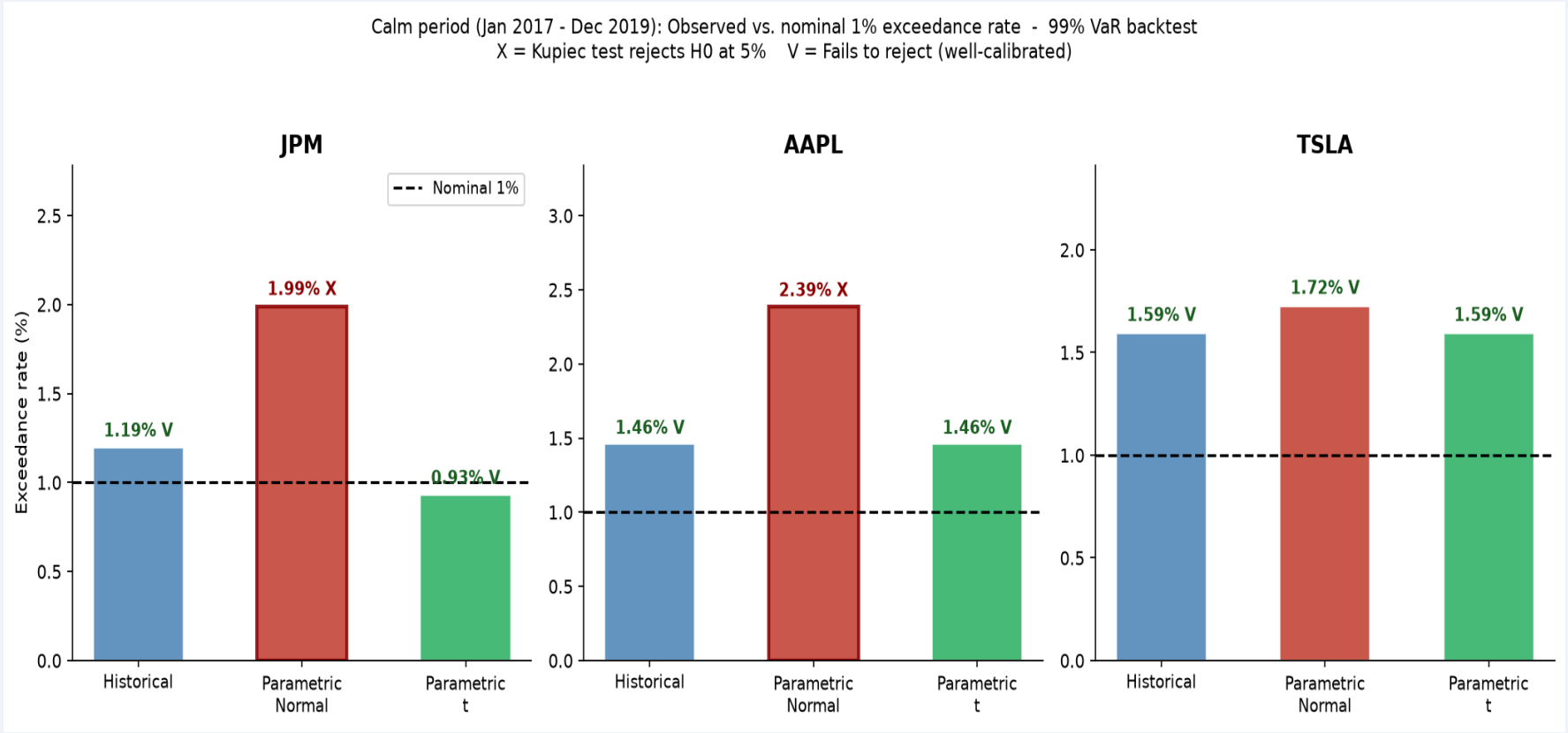
Normal bias at n = 1,000 days, α = 99% (Student-t bias < 0.4% for all tickers)

	True 99% VaR	Normal VaR bias	True 99% CVaR	Normal CVaR bias	t bias (VaR & CVaR)
JPM	4.66%	**−11.7%**	7.26%	**−34.9%**	<0.4%
AAPL	4.91%	**−13.0%**	7.31%	**−32.8%**	<0.2%
TSLA	10.02%	**−13.0%**	15.10%	**−33.7%**	<0.2%

Practical Implication:
A bank using Normal CVaR holds approximately 33% too little capital. This is systematic across all three tickers — a model mis-specification, not a data anomaly.

Backtesting — Calm Period: Normal Fails; Historical and t Pass

250-day rolling estimation window · 99% VaR · Kupiec proportion-of-failures test (H_0: observed rate = 1%)



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Backtesting — Calm Period: Normal Fails; Historical and t Pass

250-day rolling estimation window · 99% VaR · Kupiec proportion-of-failures test (H_0: observed rate = 1%)

Ticker	Method	N obs	Obs rate	Expected	LR	p-value	Reject H_0?
JPM	Historical	754	1.19%	1.0%	0.2689	0.6041	No ✓
JPM	Normal	754	**1.99%**	1.0%	5.7896	0.0161	Yes ✗
JPM	Student-t	754	0.93%	1.0%	0.0400	0.8414	No ✓
AAPL	Historical	754	1.46%	1.0%	1.4049	0.2359	No ✓
AAPL	Normal	754	**2.39%**	1.0%	10.5526	0.0012	Yes ✗
AAPL	Student-t	754	1.46%	1.0%	1.4049	0.2359	No ✓
TSLA	Historical	754	1.59%	1.0%	2.2591	0.1328	No ✓
TSLA	Normal	754	1.72%	1.0%	3.2829	0.0700	No ✓
TSLA	Student-t	754	1.59%	1.0%	2.2591	0.1328	No ✓

Note:

- For a 5% test, the critical value of $\chi^2(1)$ is 3.84. If Likelihood Ratio LR > 3.84, we reject H_0 and conclude the model is mis-calibrated.
- In calm markets, Historical and Student-t are well-calibrated. Normal generates more than ×2 the intended exceedances for AAPL.
- For TSLA, the calm-period count of 13 breaches is just below the rejection threshold. The pass is therefore inconclusive and not evidence that the Normal is well calibrated for TSLA.

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Backtesting — Stressed Period: All Models Fail (Regime Change)

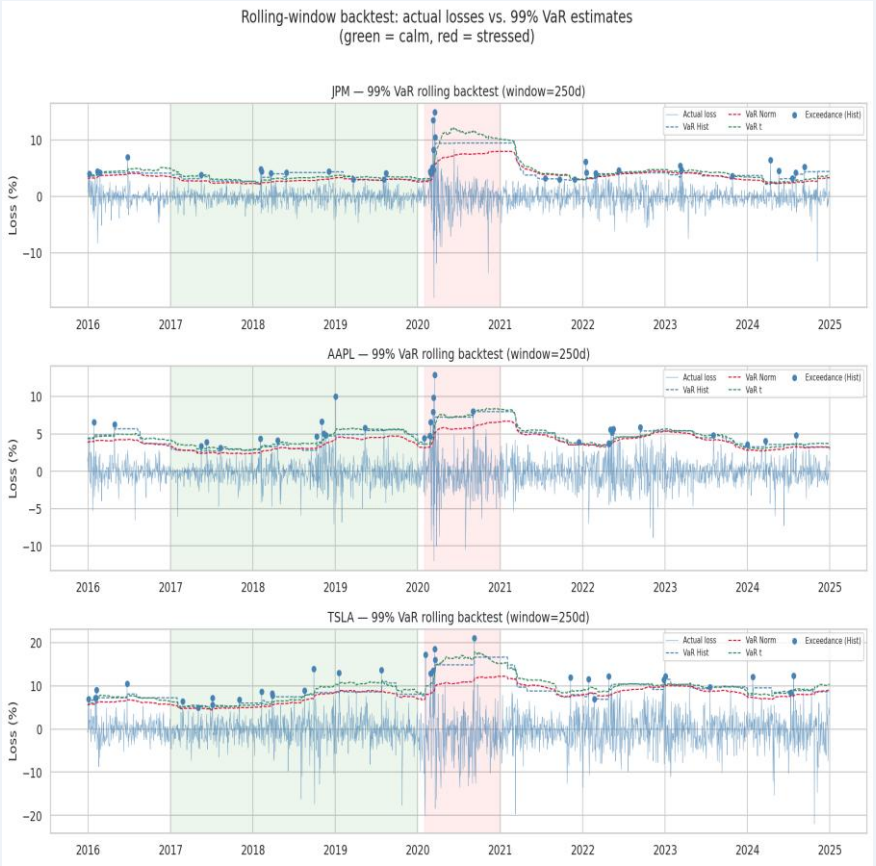
250-day rolling window · Sub-period: Feb – Dec 2020 (COVID crash) · Expected exceedance rate: 1%

Ticker	Method	N obs	Obs rate	Expected	Reject H_0?
JPM	Historical	232	4.31%	1.0%	Yes ✗
JPM	Normal	232	6.47%	1.0%	Yes ✗
JPM	Student-t	232	4.31%	1.0%	Yes ✗
AAPL	Historical	232	2.59%	1.0%	Yes ✗
AAPL	Normal	232	3.88%	1.0%	Yes ✗
AAPL	Student-t	232	2.59%	1.0%	Yes ✗
TSLA	Historical	232	2.59%	1.0%	Yes ✗
TSLA	Normal	232	3.02%	1.0%	Yes ✗
TSLA	Student-t	232	3.02%	1.0%	Yes ✗

★ Normal worst: JPM 6.47% vs 1% nominal — 6× the target rate

▶ i.i.d. assumption breaks under regime change — all estimators fail

✓ Normal remains consistently worst; failing does not mean all models are equal



Spike in exceedances during COVID crash clearly visible for all methods

Takeaway and Limitations

▶ **Recommendation:** Use the **Parametric Student-t** estimator for 99% VaR and CVaR on individual equities. It correctly accounts for heavy tails ($\nu \approx 3$–4), understates true risk by < 0.4%, converges faster than historical at short horizons ($n \leq 250$), and passes Kupiec backtests in calm markets. **Parametric Normal** understates CVaR by ~33% and should not be used for tail risk on individual equities.

01 **Calibrated DGP is a modelling choice**

The Student-t ground truth is itself a model. The true population distribution is unknown; t fit is a strong assumption.

02 **i.i.d. assumption is violated**

Volatility clustering and structural breaks mean all methods fail under regime change — as the COVID stress test confirms. A practitioner should supplement VaR with stress tests for regime scenarios.

03 **Narrow 3-ticker basket**

Findings are indicative, not universal. Results may not generalise to other assets, sectors, or portfolios.

04 **CRSP data licensing**

Raw extracts cannot be redistributed. Results are contingent on WRDS institutional access.

Thank you!